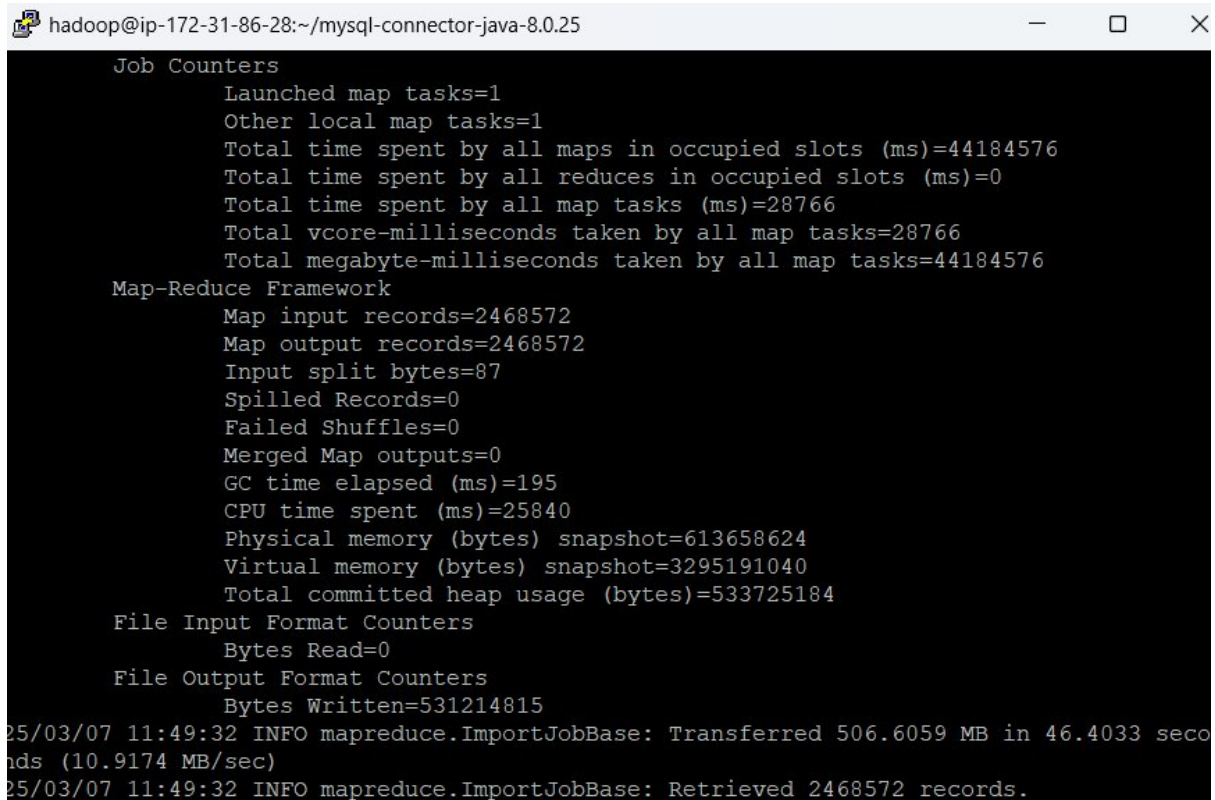


Data Ingestion from the RDS to HDFS using Sqoop

Sqoop Import command used for importing table from RDS to HDFS:

```
sqoop import \  
> --connect jdbc:mysql://upgraddetest.cyaieic9bmnf.us-east-1.rds.amazonaws.com/testdatabase \  
> --table SRC_ATM_TRANS \  
> --username student --password STUDENT123 \  
> --target-dir /user/root/bank_atm_project \  
> -m 1
```



```
hadoop@ip-172-31-86-28:~/mysql-connector-java-8.0.25  
Job Counters  
    Launched map tasks=1  
    Other local map tasks=1  
    Total time spent by all maps in occupied slots (ms)=44184576  
    Total time spent by all reduces in occupied slots (ms)=0  
    Total time spent by all map tasks (ms)=28766  
    Total vcore-milliseconds taken by all map tasks=28766  
    Total megabyte-milliseconds taken by all map tasks=44184576  
Map-Reduce Framework  
    Map input records=2468572  
    Map output records=2468572  
    Input split bytes=87  
    Spilled Records=0  
    Failed Shuffles=0  
    Merged Map outputs=0  
    GC time elapsed (ms)=195  
    CPU time spent (ms)=25840  
    Physical memory (bytes) snapshot=613658624  
    Virtual memory (bytes) snapshot=3295191040  
    Total committed heap usage (bytes)=533725184  
File Input Format Counters  
    Bytes Read=0  
File Output Format Counters  
    Bytes Written=531214815  
25/03/07 11:49:32 INFO mapreduce.ImportJobBase: Transferred 506.6059 MB in 46.4033 seconds (10.9174 MB/sec)  
25/03/07 11:49:32 INFO mapreduce.ImportJobBase: Retrieved 2468572 records.
```

Command used to see the list of imported data in HDFS:

```
hadoop fs -ls/user/root/bank_atm_project
```

Screenshot of the imported data:

hadoop fs -cat /user/root/bank_atm_project/part-m-000000(Command to view the imported data)

```
hadoop@ip-172-31-86-28:~/mysql-connector-java-8.0.25
2017,December,31,Sunday,23,Active,28,NCR,LÃfÃ, r,ÃfÃesterbrogade,8,9670,56.967,9.253,DKK,VISA,8
12,Withdrawal,,,56.962,9.258,2617467,Logstor,277.589,999,87,6,208,0.000,76,803,Clouds,broken clouds
2017,December,31,Sunday,23,Active,28,NCR,LÃfÃ, r,ÃfÃesterbrogade,8,9670,56.967,9.253,DKK,VISA,5
568,Withdrawal,,,56.962,9.258,2617467,Logstor,277.589,999,87,6,208,0.000,76,803,Clouds,broken clouds
2017,December,31,Sunday,23,Active,49,NCR,Bindslev,NÃfÃ, rrebro,18,9881,57.541,10.200,DKK,MasterCard,54
03,Withdrawal,,,57.471,10.203,2614010,Sindal,277.589,999,87,6,208,0.000,76,803,Clouds,broken clouds
2017,December,31,Sunday,23,Active,49,NCR,Bindslev,NÃfÃ, rrebro,18,9881,57.541,10.200,DKK,MasterCard,31
2,Withdrawal,,,57.471,10.203,2614010,Sindal,277.589,999,87,6,208,0.000,76,803,Clouds,broken clouds
2017,December,31,Sunday,23,Active,56,Diebold Nixdorf,KÃfÃ,ge,SÃfÃ,ndre Alle,1,4600,55.454,12.181,DKK,
Visa Dankort - on-us,5351,Withdrawal,,,55.588,12.251,2621218,Greve Kommune,280.150,992,100,3,190,0.00
0,90,804,Clouds,overcast clouds
2017,December,31,Sunday,23,Active,103,Diebold Nixdorf,Vejgaard,Hadsundvej,20,9000,57.043,9.950,DKK,Ma
stercard - on-us,5821,Withdrawal,,,57.048,9.935,2616235,NÃfÃ, rresundby,277.589,999,87,6,208,0.000,76,
803,Clouds,broken clouds
2017,December,31,Sunday,23,Active,1,NCR,NÃfÃ, stved,Farimagvej,8,4700,55.233,11.763,DKK,Visa Dankort,
8049,Withdrawal,,,55.230,11.761,2616038,Naestved,280.150,991,100,5,210,1.175,92,500,Rain,light rain
2017,December,31,Sunday,23,Active,85,Diebold Nixdorf,KÃfÃ,benhavn,Regnbuepladsen,5,1550,55.676,12.571
,DKK,MasterCard,7785,Withdrawal,,,55.676,12.566,2618425,Copenhagen,280.150,992,100,3,190,0.000,90,804
,Clouds,overcast clouds
2017,December,31,Sunday,23,Active,15,NCR,Vestre,Kastetvej,36,9000,57.053,9.905,DKK,Mastercard - on-us
,9777,Withdrawal,,,57.048,9.919,2624886,Aalborg,277.589,999,87,6,208,0.000,76,803,Clouds,broken cloud
s
2017,December,31,Sunday,23,Active,15,NCR,Vestre,Kastetvej,36,9000,57.053,9.905,DKK,MasterCard,5531,Wi
thdrawal,,,57.048,9.919,2624886,Aalborg,277.589,999,87,6,208,0.000,76,803,Clouds,broken clouds
2017,December,31,Sunday,23,Active,34,NCR,Skipperen,Vestre Alle,2,9000,57.034,9.908,DKK,MasterCard,832
8,Withdrawal,,,57.048,9.919,2624886,Aalborg,277.589,999,87,6,208,0.000,76,803,Clouds,broken clouds
2017,December,31,Sunday,23,Active,20,NCR,Bispensgade,Bispensgade,35,9800,57.453,9.996,DKK,MasterCard,
5023,Withdrawal,,,57.464,9.982,2620214,Hjorring,278.589,998,99,8,210,0.000,44,802,Clouds,scattered cl
ouds
2017,December,31,Sunday,23,Active,34,NCR,Skipperen,Vestre Alle,2,9000,57.034,9.908,DKK,MasterCard,147
,Withdrawal,,,57.048,9.919,2624886,Aalborg,277.589,999,87,6,208,0.000,76,803,Clouds,broken clouds
2017,December,31,Sunday,23,Active,70,Diebold Nixdorf,Holstebro,Hostrupsvej,6,7500,56.373,8.625,DKK,Ma
sterCard,5666,Withdrawal,,,56.360,8.616,2620046,Holstebro,280.150,988,93,4,210,0.000,92,804,Clouds,ov
ercast clouds
2017,December,31,Sunday,23,Active,49,NCR,Bindslev,NÃfÃ, rrebro,18,9881,57.541,10.200,DKK,MasterCard,78
86,Withdrawal,,,57.471,10.203,2614010,Sindal,277.589,999,87,6,208,0.000,76,803,Clouds,broken clouds
2017,December,31,Sunday,23,Inactive,12,NCR,ÃfÃesterÃfÃ¥ Duus,ÃfÃesterÃfÃ¥,12,9000,57.049,9.922,DKK,M
astercard - on-us,2436,Withdrawal,,,57.048,9.919,2624886,Aalborg,277.589,999,87,6,208,0.000,76,803,Cl
ouds,broken clouds
2017,December,31,Sunday,23,Inactive,12,NCR,ÃfÃesterÃfÃ¥ Duus,ÃfÃesterÃfÃ¥,12,9000,57.049,9.922,DKK,M
astercard - on-us,8519,Withdrawal,,,57.048,9.919,2624886,Aalborg,277.589,999,87,6,208,0.000,76,803,Cl
ouds,broken clouds
2017,December,31,Sunday,23,Active,10,NCR,NÃfÃ, rresundby,Torvet,6,9400,57.059,9.922,DKK,Mastercard - o
n-us,5286,Withdrawal,,,57.048,9.919,2624886,Aalborg,277.589,999,87,6,208,0.000,76,803,Clouds,broken c
louds
```